b. Chemical properties

i. Reaction with metals

Generally metals do not react with bases. Metals like aluminium and zinc react with bases like sodium hydroxide forming aluminates and release hydrogen.

Aluminum + Sodium hydroxide + Water → Sodium aluminate + Hydrogen

$$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2\uparrow$$

ii. Reaction with non-metal oxides

All bases react with non metallic oxides to form salt and water. For example, sodium hydroxide reacts with carbon dioxide to form sodium carbonate.

Sodium hydroxide + Carbon dioxide → Sodium carbonate + Water

$$2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$$

iii. Reaction with ammonium salts

Bases react with ammonium salts to form metal salts, ammonia gas and water.

Sodium hydroxide + Ammonium chloride → Sodium chloride + Ammonia + Water

$$NaOH + NH_4Cl \rightarrow NaCl + NH_3\uparrow + H_2O$$

Though acids and bases have some unique properties there are certain similarities between them. Some of them are given below.

- They are corrosive in nature.
- They undergo ionization in aqueous solution.
- They conduct electricity in aqueous solution.
- They undergo neutralization reaction.

Some of the differences between acids and bases are given in Table 14.3.

Table 14.3 Difference between acids and bases

Acids	Bases
They produce H^+ ions in water.	They produce OH^- ions in water.
They are sour in taste.	They are bitter in taste.
Few acids are in solid state.	Most of the bases are in solid state.
Acids turn blue litmus paper red.	Bases turn red litmus paper blue.

14.2.2 Uses of Bases

i) Potassium hydroxide is used to make bathing soaps.
ii) Sodium hydroxide is used to make washing soaps.
iii) Sodium hydroxide is also used in paper industries, textile industries and in the preparation of medicines.
iv) Calcium hydroxide is used for white washing.
v) Aluminum hydroxide and magnesium hydroxides are used in antacids to cure acidity problems.
vi) Ammonium hydroxide is used to manufacture fertilizers, nylon, plastics and rubber.



Figure 14.4 Uses of bases in daily life

14.3 Neutralisation Reaction

When neutrality is achieved between two different chemical substances with different chemical properties through a reaction then it is called neutralization in chemistry. Thus neutralization is a chemical reaction in which

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